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Studierichting: Informatica

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$$\begin{aligned} \int \frac{1+2\sin x}{1+\cos x+\sin x} dx &\stackrel{t=\tan(\frac{x}{2})}{=} \int \frac{1+\frac{4t}{1+t^2}}{1+\frac{1-t^2}{1+t^2}+\frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= 2 \int \frac{1+\frac{4t}{1+t^2}}{1+t^2+1-t^2+2t} dt = \int \frac{1+t^2+4t}{(t+1)(t^2+1)} \\ &\Rightarrow \frac{t^2+4t+1}{(t+1)(t^2+1)} = \frac{A}{t+1} + \frac{Bx+C}{t^2+1} \\ &\Rightarrow A = \frac{1-4+1}{1+1} = -1 \\ &\Rightarrow B(i)+C = \frac{-1+4i+1}{i+1} = 2+2i \Rightarrow B=2, C=2 \\ &= -\int \frac{1}{t+1} dt + 2 \int \frac{1+t}{t^2+1} dt = -\ln|t+1| + 2\text{Bgtan } t + \ln|t^2+1| + c \\ &\stackrel{t=\tan(\frac{x}{2})}{=} -\ln\left|\tan\left(\frac{x}{2}\right)+1\right| + \ln\left|\tan^2\left(\frac{x}{2}\right)+1\right| + x + c \end{aligned}$$

References